UNIVERSITY PROJECT | IoT + MOBILE

SafeBand

A smart, screenless safety wearable that detects danger and calls for help automatically.

Detect. Confirm. Alert.

THE PROBLEM

When danger strikes, reaching for a phone is not an option

01

Hard to reach

Phones are in bags or pockets, and every second counts.

02

Hard to call

Victims are often too frightened or restrained to dial or speak.

03

Too manual

Most safety apps need several taps before any help is sent.

Women need protection that works even when they cannot act.

OUR SOLUTION

One band. Four ways to get help.

01

Auto detection

Heart rate and motion sensors spot panic and struggle without any input.

02

Triple-press SOS

Press the button three times for an instant, silent alert.

03

Live location alert

SMS, call and a live tracking link go to family and police.

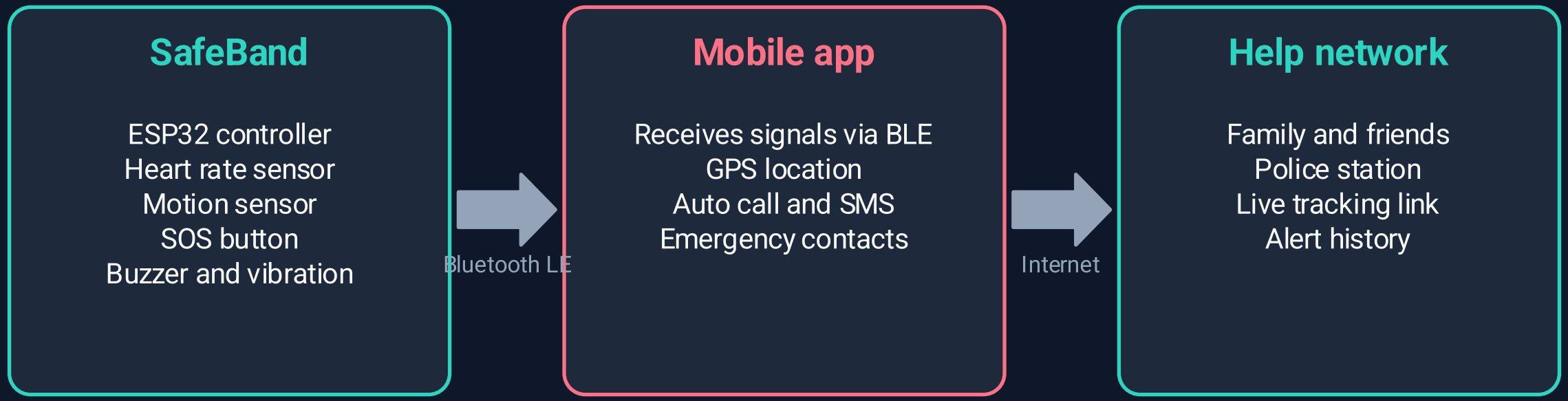
04

Siren and vibration

A loud buzzer draws attention, and vibration warns the wearer.

SYSTEM ARCHITECTURE

How the pieces connect



The band needs no SIM or GPS, which keeps it small, cheap and light on battery.

HOW IT WORKS

From sensing to rescue in four steps

1

Sense

Heart rate and motion are measured continuously.

2

Analyse

High heart rate plus violent motion signals possible danger.

3

Confirm

The band vibrates for 10 seconds. Tap to cancel a false alarm.

4

Alert

Call, SMS and live location are sent. The siren starts.

MODERN UPGRADES

What makes SafeBand next generation

On-device AI

A TinyML model tells real danger from running or dancing, cutting false alarms.

Voice trigger

Say a secret keyword to send a silent alert with no button press.

Evidence capture

Short audio is recorded and shared securely with the alert.

Encrypted alerts

Location and messages are end-to-end encrypted for privacy.

Cloud dashboard

Family sees live tracking and alert history in one place.

Built to last

Water resistant, wireless charging, low-power Bluetooth for days of use.

ROADMAP AND IMPACT

Built in three phases

Phase 1

SOS button, Bluetooth and app alerts.

Phase 2

Motion detection and heart rate analysis.

Phase 3

AI model, voice trigger and cloud dashboard.

Safety by design: we use loud sirens and vibration, not electric shock, so the band protects without risking harm.

Thank you.